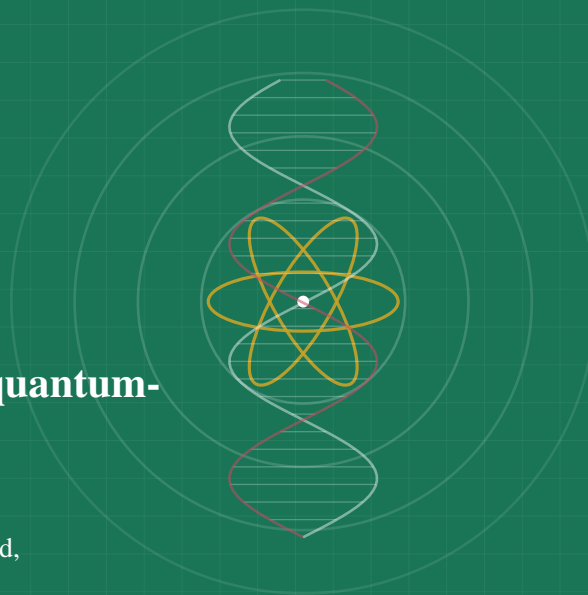


Beyond the Reach of the Body?

Inventing mathematics for the nano-, quantum- and cellular scale

If mathematics is embodied, is it capped by our cognition?

A precise answer, five historical escapes, a repeatable method, and an audit of what is actually missing.



LEVER 1

Extend the body
instruments & sensorimotor loops

LEVER 2

Re-ground the abstraction
diagrammatic calculi

LEVER 3

Externalise the bookkeeping
machines & proof assistants

The correction this dossier turns on. Lakoff & Núñez's thesis constrains the *route*, not the *destination*. Every mathematical concept must be **reachable** from grounded experience by a chain of metaphors and blends. It need not be **picturable**.

Nobody visualises an infinite-dimensional Hilbert space, a non-commutative torus, or an ∞ -groupoid. All three are computed with, daily, without error — because mathematics re-grounds in a *different* body of experience when vision runs out.

What is in this dossier

- a line-by-line audit of a second AI's answer to the same question
- a five-move method you can actually run
- an honest ledger of quantum biology's evidence
- five documented escapes past intuition, with dates
- what mathematics already exists for the cell, by name and date
- the bottleneck that is *not* mathematics

Every historical claim carries a name and a year.

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Provenance and conventions

This dossier answers a question put by the reader and audits a long response to the same question supplied from another AI assistant. **Personal names appearing in that pasted text — two students — have been removed throughout.** The pasted text is referred to as “the pasted text” and is quoted only in short, attributed fragments.

✓ correct ~ partly correct or imprecise ✗ incorrect or misleading · PURPLE SMALL CAPS = a metaphor · **green** a move you can run **red** a caution or correction

1 The question, and the premise checked

The question as posed

Since our mathematics came mostly from human physical realities of thought, our mathematics is limited to our cognitive limitations. In that case, how do we invent new mathematics to explore the new worlds of nano/quantum-scale biological systems inside cells, molecules, atoms? What can we do about that?

It is the natural inference to draw from Part VI — and it is the one Lakoff & Núñez themselves are careful not to draw. Separating the part that is correct from the part that does not follow is what tells you what to do.

1.1 Two claims that get conflated

CONCEIVABILITY CONSTRAINT

You must be able to *build* the concept out of grounded pieces, via metaphor and blending.

TRUE. This is the book's actual thesis.

It constrains the **route**.

≠

IMAGINABILITY CONSTRAINT

You must be able to *picture* the result.

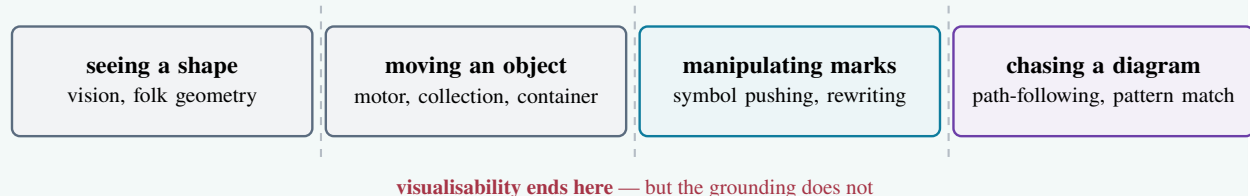
FALSE — and nothing in Part VI asserts it.

It would constrain the **destination**. It does not exist.

The evidence against the second is overwhelming, and you already hold it. Nobody pictures an infinite-dimensional separable Hilbert space. Nobody pictures a non-commutative torus. Nobody pictures an ∞ -groupoid. Mathematicians compute with all three fluently and without error.

How the escape actually works: substituting the source domain

When vision and folk physics run out, mathematics does not become disembodied. It **re-grounds in a different body of experience**:



Hand, eye and motor cortex are still doing the work in the right-hand two columns. They are simply not doing *spatial imagery*. This distinction is not a technicality: it tells you exactly which lever to pull. If the obstacle were imagination, you would be stuck. Because the obstacle is grounding, you can **build a new grounding**.

And the book says so, one page before the Part VI you read:

“*Mathematics is creative and open-ended. By virtue of the use of conceptual metaphors and conceptual blends, present mathematics can be extended to create new forms by importing structure from one branch to another and by fusing mathematical ideas from different branches.*

BOOK P. 379 / PDF P. 398

1.2 The defensible version of your worry

What survives the correction — and it is not nothing

The right conclusion is not “*there is a ceiling*.” It is:

*Unintuitive mathematics is slower to invent, harder to teach,
and more error-prone.*

That is a real and expensive cost. Non-Euclidean geometry took roughly two millennia, and it took that long largely because the parallel postulate *felt* necessary — which is an embodiment effect. But a tax on the rate of discovery is not a wall across the road.

2 Audit of the second AI’s response

You supplied a long answer to the same question from another assistant. Personal names have been removed from it here. It is a good answer — substantially correct, well organised, and it reaches the right headline conclusion. What follows is a precise audit: what it gets right, what it gets right for the wrong reason, and what it leaves out. The omissions are the actionable part.

2.1 Scorecard

Claim in the pasted text		Verdict
Mathematics began from human-scale experience (counting, land, motion, circles)	✓	Correct, and exactly Part VI's grounding metaphors.
"Mathematics is not trapped inside human-scale physics"	✓	The right headline. Same conclusion this dossier reaches.
The need is for <i>new languages and representations</i> , not necessarily new mathematics	✓	Well judged, and more precise than the question presumed.
$i = \sqrt{-1}$ offered as an example of escaping intuition	✗	Self-undermining. Case Study 3 is the book's flagship demonstration that i is grounded — in mental rotation. See §2.2.
"Formal symbolic reasoning can exceed human intuition"	~	Right conclusion, wrong mechanism . Symbol pushing is not disembodied; it is re-grounded. The difference changes what you should do. See §2.3.
A protein is simultaneously structure, dynamics, landscape, stochastic object, information carrier, network node, evolutionary object	✓	Genuinely useful framing, and the strongest original observation in the piece.
Scale span 10^{-10} m to 1 m; 10^{-15} s upward	✓	Correct. Made precise in §5, with which mathematics covers which band.
Ingredients already exist: stochastic calculus, graphs, dynamical systems, information theory, topology, statistical mechanics, quantum mechanics	~	True but unsourced and generic . Every one of these has a specific, dated result that did the actual work. Named in §5 and the appendix.
Category theory: relations and transformations as primary	✓	Correct, and the right instinct. This is Move 2 of the method.
"Property of X" must become "Property of X Context C"	✓✓	The best idea in the piece. It already has a precise mathematical home the text does not name. See §2.4.
Emergence: micro rules → macro behaviour needs better mathematics	✓	Correct; this is a genuine open problem, not a gap in anyone's education.
Cells are open, far-from-equilibrium systems	✓	Correct, and the sharpest real gap. Named results exist; a general variational principle does not.
Open quantum systems, not textbook isolated QM	✓	Correct. But the evidence base needs grading — see §6.
Quantum biology presented without grading the evidence	✗	Photosynthetic "long-lived electronic coherence" was substantially revised after 2007. A student given this uncritically will be misled.
AI: data → pattern → conjecture → human+machine proof	✓	Correct, and there is now real evidence for it. Named in §7.
"AI may discover $F(X) = \text{constant}$ "	~	Plausible, but offered with no example. Two real ones exist. §7.
The ten-step research loop	~	Sound but generic — it would fit any field. It lacks the one step that is specific to the embodiment problem: audit the failing metaphor .

Table 1. ✓ correct ~ partly correct or imprecise ✗ incorrect or misleading ✓✓ correct and undervalued by its own author.

2.2 The $i = \sqrt{-1}$ example is self-undermining

The pasted text writes: "*There is no ordinary physical pile containing i apples*" — and offers this as evidence that mathematics escapes intuition.

Why this is exactly backwards

The observation is true and the conclusion drawn from it is wrong. **Case Study 3 of the very book in question spends thirteen pages showing that i is grounded** — not in collections of apples, but in **MULTIPLICATION BY -1 IS ROTATION BY 180°** , extended to **ROTATION BY 90° IS MULTIPLICATION BY i** .

fails the collection grounding
no pile of i apples

fails the magnitude grounding
not linearly ordered

succeeds on the rotation grounding
mental rotation, R_{-1}

therefore i is grounded,
and the complex plane is derived

i is not an example of mathematics leaving the body. **It is the book's cleanest example of mathematics finding a different part of the body to stand on** — which is the whole lesson, and the whole method. Getting this backwards costs you the technique.

2.3 Right conclusion, wrong mechanism — and why that matters

The pasted text's mechanism

"Formal symbolic reasoning can exceed human intuition."

Implied prescription: **be more formal**. Retreat into symbols and away from meaning.

Problem: this is Benjamin Peirce's position — proof without understanding. Part VI is a book-length argument against it.

The mechanism that actually operates

Symbol manipulation is *itself* grounded — in the hand and eye moving marks around.

Implied prescription: **build a better calculus to manipulate**.

This is a constructive instruction. It tells you to invent Feynman diagrams, string diagrams, the ZX-calculus — see Move 3.

Figure 1. The same conclusion, reached two ways, yielding two completely different instructions. Only the right-hand one is actionable.

2.4 The best idea in the piece already has a name

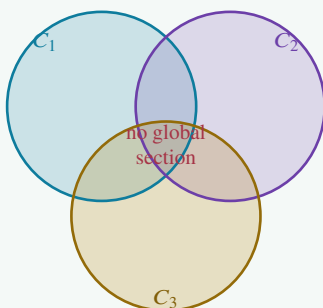
The pasted text proposes, almost apologetically (*"That sounds philosophical"*):

$$\text{Property of } X \quad \longrightarrow \quad \text{Property of } X \mid \text{Context } C$$

and notes that a protein behaves differently depending on location, binding partners, phosphorylation state, cell type, temperature, pH and expression state.

This is not philosophy — it is contextuality, and it is a theorem

The mathematics of "properties that are locally well-defined but have no consistent global assignment" exists and is precise. **Abramsky & Brandenburger (2011)** showed that quantum contextuality and non-locality are exactly the obstruction to gluing local sections of a **presheaf** into a global section — a cohomological obstruction.



Each context C_i admits a perfectly consistent local assignment of properties. **No single global assignment restricts correctly to all of them.** That is contextuality, and it is measured by a sheaf cohomology class.

Why this matters for your question: a protein whose properties are well-defined per assay but not jointly is *formally the same situation*. The mathematics is already built. It has simply not been carried across to molecular biology in earnest.

This is a live, tractable research direction — and it is the single most valuable thing in the text you pasted, which is why it deserved a name and a citation rather than an apology.

2.5 What the pasted text leaves out entirely

Omission	Why it is the actionable part
Instruments as mathematics-generators	The pipeline given is Experiment → Data → AI → Conjecture. It never notes that <i>instruments create new intuitions</i> . The STM (1981) and AFM (1986) put atoms into a sensorimotor loop, and new mathematics followed within a decade. This is Lever 1, and it is missing.
Diagrammatic re-grounding	No Feynman diagrams, no Penrose notation, no string diagrams, no ZX-calculus. This is historically the highest-yield move for exactly this problem, and it is absent. This is Lever 2.
Names and dates	“Some of the ingredients already exist” — which, by whom, when? Feinberg’s deficiency theorems, Hopfield’s kinetic proofreading, Jarzynski and Crooks, HEOM, Abramsky–Brandenburger. Without these a student cannot start reading tomorrow.
The measurement bottleneck	For most of molecular and cellular biology <i>right now</i> , the binding constraint is not mathematics — it is that we cannot watch single molecules inside a living cell at sufficient spatial <i>and</i> temporal resolution simultaneously. Inventing mathematics for data you cannot take is the wrong order of operations.
An operational test for “we need new mathematics”	The ten-step loop says “identify the failure” but not how. The metaphor audit (Move 1) is the missing instrument.

A note on style, offered for your students’ sake

The pasted text makes heavy use of boxed displays such as **Matter + Information + Probability + Geometry + Life**. This is effective rhetoric and there is nothing wrong with it as motivation. But it is worth teaching explicitly that **a box around a list of nouns is not a theorem, and “+” between fields is not an operation**. Part VI’s whole discipline is the refusal to let notation stand in for structure. A student who learns to write the box before they can write the definition has learned the wrong lesson from Euler.

3 Five escapes past intuition, with dates

This is not speculation. Each of the following is a documented case of mathematics deliberately built past embodied intuition. Each followed **the same two-step pattern**, and recognising that pattern is what makes the method repeatable.

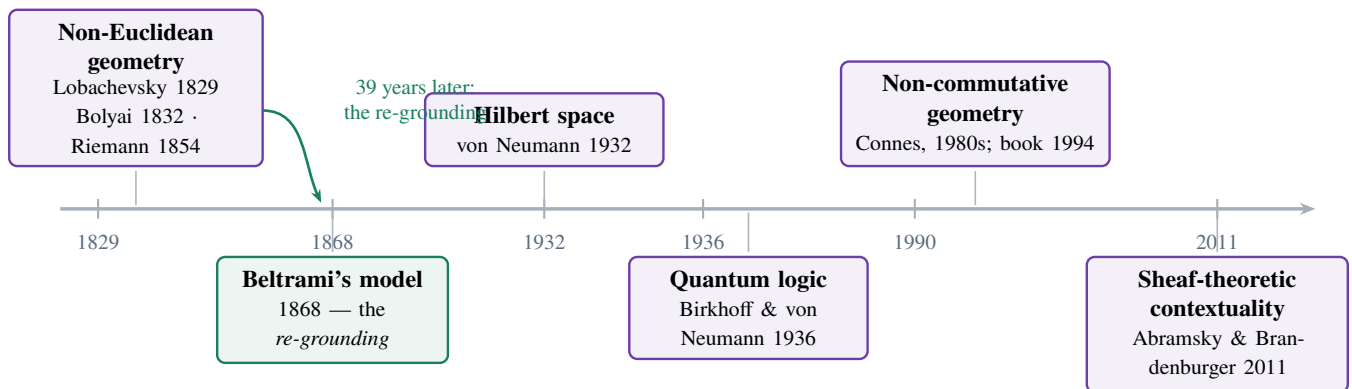


Figure 2. Chronology of the five escapes. Note the gap between the symbolic escape (1829) and its re-grounding in a drawable model (1868). That gap is the cost of unintuitive mathematics, measured in decades.

3.1 The two-step pattern

STEP 1 — Symbolic escape

Negate or delete the assumption that *feels* necessary. Derive consequences. Check for contradiction.

You are now computing with something you cannot picture. This is uncomfortable, slow, and historically takes decades to be accepted.

STEP 2 — Re-grounding

Build a *new blend* — a model, a diagram, a calculus — that makes the structure manipulable by hand and eye again.

Beltrami's disk. Feynman's diagrams. The ZX-calculus. This is when the mathematics becomes *usable* rather than merely valid.

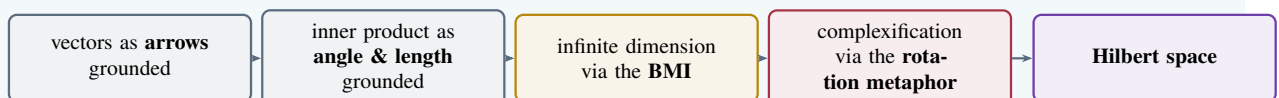
3.2 The cases, one paragraph each

1. Non-Euclidean geometry (Lobachevsky 1829 · Bolyai 1832 · Riemann 1854 · Beltrami 1868)

The parallel postulate felt *necessary* precisely because it matched embodied spatial experience. The escape was symbolic: negate it, derive consequences, find no contradiction. Only **39 years later** did Beltrami build the pseudosphere model, and then Klein and Poincaré their disk models — new blends that re-grounded the abstract structure in something you could draw.

2. Hilbert space (von Neumann, *Mathematische Grundlagen der Quantenmechanik*, 1932)

An electron's state is a ray in an infinite-dimensional complex projective space; observables are self-adjoint operators; measurement is a projection. Not visualisable. But look at the assembly:



The machinery of Part VI is what built quantum mechanics' state space. The BMI of Chapter 9 and the 90° rotation metaphor of Case Study 3 are load-bearing components of the formalism you would use to model an electron in a protein.

3. Quantum logic (Birkhoff & von Neumann, *Annals of Mathematics*, 1936)

The lattice of propositions about a quantum system is **non-distributive**. Distributivity feels necessary because of the container/collection metaphor: if the marble is in box *A* or box *B*, it is in one of them. Delete that, and you have a logic for systems where it is not so.

4. Non-commutative geometry (Connes, 1980s; *Noncommutative Geometry*, 1994)

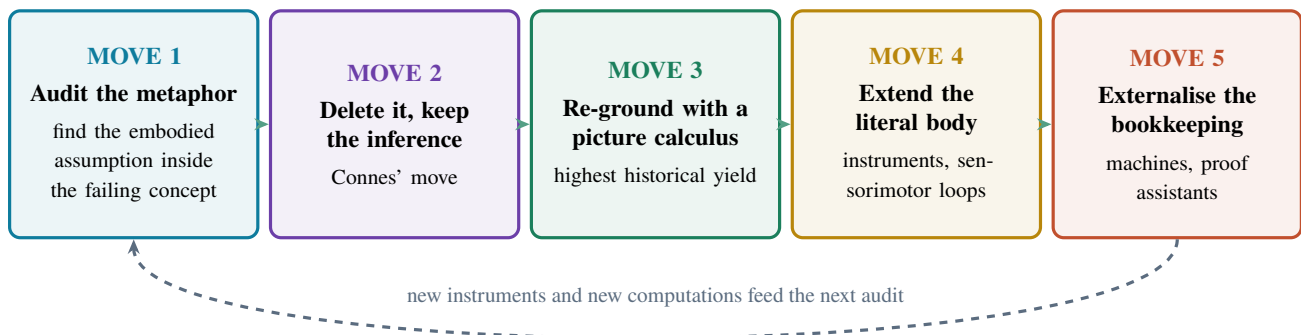
The sharpest case for your question. Classical geometry says a space is a set of points, and functions on it commute because you evaluate at a point. Quantum observables do not commute. Connes' move: *stop insisting on points*. Take the algebra as primary, drop commutativity, and see what "geometry" survives — you get a geometry with no points, which is what a quantum "space" requires.



5. Sheaf-theoretic contextuality (Abramsky & Brandenburger, *New J. Phys.*, 2011)

Quantum contextuality and non-locality turn out to be exactly the obstruction to gluing local sections of a presheaf into a global section. A piece of mid-century algebraic topology became the right language for a phenomenon with no classical picture at all — and, as §2.4 argued, plausibly the right language for context-dependent molecular identity too.

4 The method: five moves you can actually run



MOVE 1 — Audit the metaphor that is failing

Take a concept that is breaking and find the embodied assumption inside it. This is the step the pasted text's ten-point loop does not have, and it is the one specific to your problem.

Worked example, from your own target domain. The statement “*species X has concentration [X]*” presupposes the **CONTINUOUS SUBSTANCE** metaphor — stuff you could pour. Inside a bacterial cell with twelve copies of a transcription factor, that metaphor is simply false.



The stochastic simulation algorithm did not arrive because someone wanted more mathematics. It arrived because someone noticed a metaphor had stopped applying.

MOVE 2 — Delete the offending assumption, keep the inference

Connes' move, generalised. Candidate assumptions to attack for nanoscale biology, with the mathematics each deletion opens:

Embodied assumption to delete	What opens up
“a system has a definite state”	density matrices; open-system dynamics; Lindblad 1976
“a space is made of points”	non-commutative geometry; pointless topology, locales
“wholes are unions of parts”	entanglement, non-separability; operads
“properties are simultaneously determinate”	contextuality; quantum logic; sheaf cohomology
“there is a global clock”	in a cell there is not — molecular events share no time coordinate
“identity persists”	indistinguishability; anyons; and, in biology, the fact that a “protein X” is a population, not an individual

MOVE 3 — Re-ground with a new pictorial calculus (*the highest-yield move in the history of this problem*)

Once you hold an unvisualisable abstract structure, invent a **diagrammatic** calculus that makes it manipulable by hand and eye.

Feynman diagrams
1948

Penrose notation
1971

string diagrams
monoidal categories

ZX-calculus
Coecke & Duncan 2008

tensor networks
MPS, PEPS

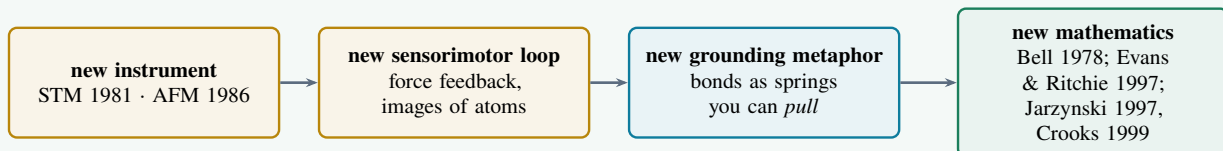
The ZX-calculus deserves dwelling on: it turns Hilbert-space computation into **rewriting pictures by local rules**, and it is provably complete for qubit quantum mechanics. You are not visualising the state space. You are re-grounding the computation in pattern-matching on diagrams — something bodies are extremely good at.

This is the most direct available answer to “our cognition is limited”: build a new body for the mathematics to live in.

MOVE 4 — Extend the literal body with instruments

The move people forget, and the most concrete one.

The reason we have *any* nanoscale intuition is that Binnig and Rohrer’s scanning tunnelling microscope (1981; Nobel Prize 1986) and the atomic force microscope (Binnig, Quate & Gerber, 1986) put atoms into a **sensorimotor loop** — force feedback and images for objects 10^{-9} m across. New mathematics followed directly:



Single-molecule pulling produced the Bell–Evans–Ritchie theory of forced unbinding, and then the non-equilibrium work relations that let you extract *equilibrium* free energies from *irreversible* pulling experiments.

Lakoff & Núñez’s own theory predicts this: new embodied experience → new grounding metaphors → new mathematics. The answer is not to escape embodiment. It is to **extend the body**.

MOVE 5 — Externalise the bookkeeping

Genuinely new in this century, and the most direct assault on the original worry.

Instrument	Proof of concept
Proof assistants	The Liquid Tensor Experiment : Scholze posed it in December 2020 because he did not fully trust his own proof; completed in Lean in July 2022. Mathematics too large to hold in one head, verified anyway.
Learned structure	AlphaFold2 (Jumper et al., <i>Nature</i> , 2021) did not give humans folding intuition. It found structure in a space no human can traverse.

But be honest about what this buys

It changes what **verification** costs. It does not by itself change what **understanding** means. A Lean proof with no idea analysis behind it is Benjamin Peirce all over again — “*we have proved it, and therefore we know it must be truth.*” Part VI is precisely the argument that this is not enough. Move 5 without Move 3 gives you correctness without comprehension.

5 The target: what exists, what is missing

Here I want to be realistic rather than inspiring. The pasted text is right that biology spans extraordinary scales. What it does not do is say *which mathematics already covers which band*, and *where the coverage actually stops*.

5.1 The scale ladder, with coverage

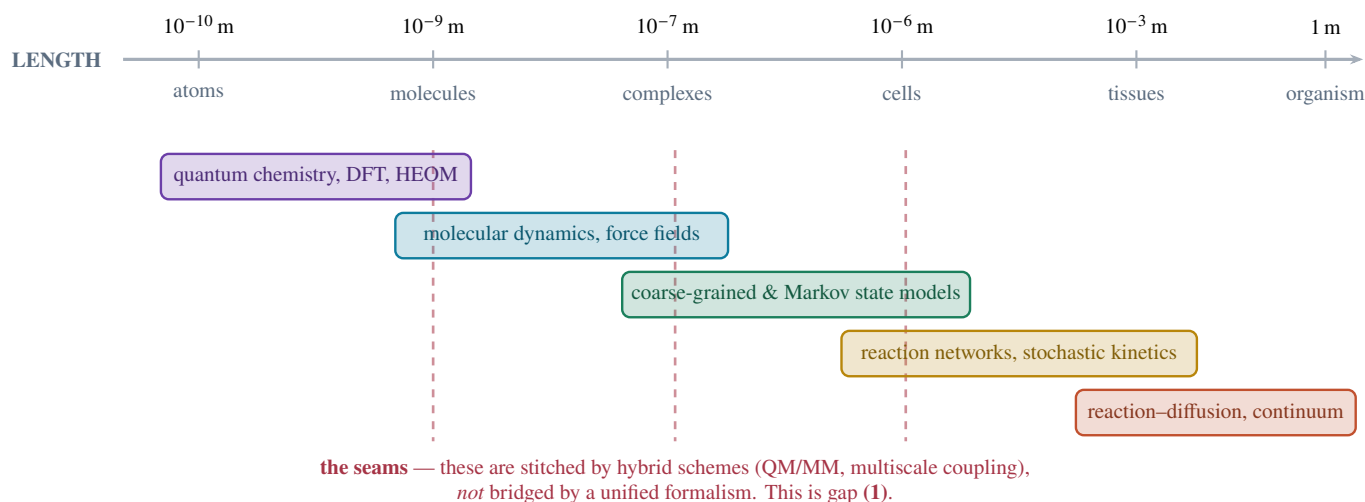


Figure 3. Which mathematics covers which length scale. The overlaps are real; the *transitions* are where the formalism changes character entirely, and nothing carries across them automatically.

5.2 What already exists and works — by name and date

The pasted text says “some of the ingredients already exist.” Here are the specific results that did the actual work, so a student can start reading tomorrow.

Problem	The mathematics that solved it
Network topology $\Rightarrow$ dynamics, <i>without</i> knowing rate constants	Deficiency-zero and deficiency-one theorems — Horn & Jackson 1972; Feinberg, 1972 onwards. A structural property of the reaction graph guarantees existence, uniqueness and stability of equilibria. This is the outstanding example of mathematics tailored to molecular systems.
Why DNA replication is more accurate than binding energies allow	Kinetic proofreading — Hopfield 1974 (PNAS); Ninio 1975. Error correction bought with energy dissipation.
Free energies from irreversible pulling	Jarzynski equality 1997 (PRL); Crooks fluctuation theorem 1999 (PRE).
Cost of precision in a driven system	Thermodynamic uncertainty relation — Barato & Seifert 2015 (PRL).
Small copy numbers, discrete molecules	Chemical master equation ; Gillespie stochastic simulation algorithm, 1976–77.
Protein conformational dynamics from simulation	Markov state models ; persistent homology (Edelsbrunner, Letscher & Zomorodian 2002; Carlsson 2009) applied to folding trajectories and to cryo-EM heterogeneity.
Strongly coupled, non-Markovian environments	HEOM — Tanimura & Kubo 1989. Numerically exact; expensive.
Composing biochemical modules	Applied category theory — Baez & Pollard, <i>A compositional framework for reaction networks</i> , 2017; decorated cospans.

5.3 Where the mathematics genuinely runs out

GAP 1 — Scale bridging

QM/MM hybrids are *stitching*, not a unified formalism.

There is no general theory of **what survives coarse-graining** for open, driven systems.

The renormalisation group is the nearest relative — and it works beautifully at equilibrium critical points, which is not where cells live.

GAP 2 — A variational principle far from equilibrium

At equilibrium, *minimise free energy* organises everything.

There is no accepted analogue for driven systems.

Maximum entropy production and related proposals remain contested. This is a genuine open problem, not a gap in anyone's education.

GAP 3 — Non-Markovian open systems

No clean general theory.

HEOM is powerful but is essentially numerically exact brute force.

Precisely the regime a chromophore in a warm protein actually occupies — which is why quantum biology is mathematically hard as well as experimentally hard.

The bottleneck that is not mathematics — read this before choosing a problem

For most of molecular and cellular biology **right now**, the binding constraint is **measurement**, not mathematics. We cannot yet watch single molecules inside a living cell with sufficient spatial *and* temporal resolution simultaneously. Inventing mathematics for data you cannot take is the wrong order of operations.

Which loops straight back to **Move 4**. If the constraint is measurement, then the highest-value contribution available is often *an instrument* — and the instrument will generate the mathematics, as the STM and AFM did.

6 Quantum biology: an honest ledger

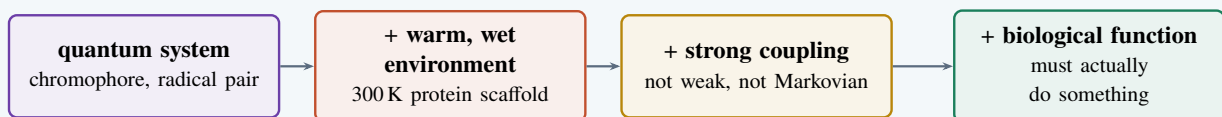
This field has a credibility problem, and the pasted text presents it without grading. A student handed that version will be misled — not about the mathematics, but about which phenomena are actually there to be modelled. Here is the spread.

Status	Phenomenon	Evidence
SOLID	Hydrogen tunnelling in enzyme catalysis	Well established and uncontroversial. Kinetic isotope effects far larger than semiclassical theory permits. Model with standard rate theory plus tunnelling corrections.
STRONG, not closed	Radical-pair magnetoreception in migratory birds	Proposed by Schulten et al. 1978; cryptochrome identified as the candidate protein. Hore & Mouritsen 2016 (<i>Annu. Rev. Biophys.</i>) is the standard review; Xu et al., <i>Nature</i> 2021, showed magnetic sensitivity of robin cryptochrome-4 <i>in vitro</i> . Behavioural link is strong; the <i>in vivo</i> chain is not yet closed.
SUBSTANTIAL REVISITED	“Long-lived electronic coherence” in photosynthetic light harvesting	Engel et al., <i>Nature</i> , 2007 was the celebrated result. Subsequent work attributes much of the observed long-lived beating to vibrational , not electronic, coherence. The strong original claims did not survive. Some coherent transport physics remains; the headline does not.
WIDELY REJECTED	Orchestrated objective reduction in microtubules (Penrose–Hameroff)	Tegmark, <i>Phys. Rev. E</i> , 2000, estimated decoherence times in the brain many orders of magnitude too short. This is the standard objection and it has not been answered.

Table 3. Grading matters practically: it tells you which phenomena will reward mathematical effort and which will waste it.

The genuine mathematical problem underneath all four

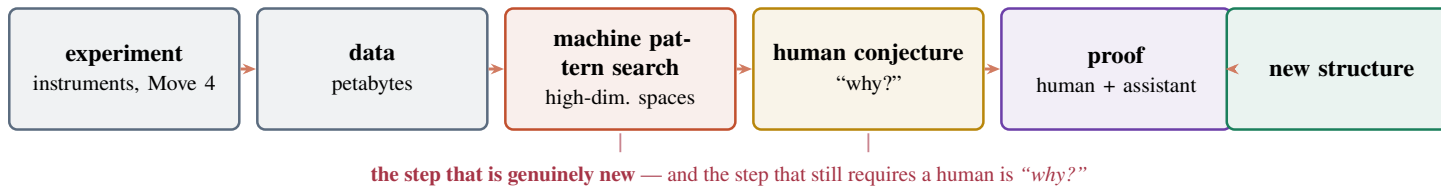
Whatever you believe about any individual phenomenon, the *mathematical* situation is the same and it is hard:



No clean general theory exists for this regime. Lindblad requires weak coupling and Markovianity. HEOM is numerically exact but scales badly. **This is Gap 3, and it is where a mathematician could genuinely contribute.**

7 AI as the new instrument — what is actually established

The pasted text’s most forward-looking claim is that AI may discover invariants humans would miss, and that mathematics will become a human–machine partnership. This is correct, and unlike the rest of that section it can be supported with specific results rather than expectation.



Claim in the pasted text	The actual evidence for it
“AI may discover that data follows a previously unnoticed invariant”	Davies et al., <i>Nature</i>, 2021 — “Advancing mathematics by guiding human intuition with AI”. Machine-learned attribution pointed to two genuine results: a new relationship between knot invariants (algebraic and geometric), and a conjecture in representation theory on Kazhdan–Lusztig polynomials. Both were then proved by humans. This is exactly the loop above, and it has already run. Symbolic regression — e.g. AI Feynman (Udrescu & Tegmark, 2020): recovering closed-form physical laws from data. Works well on clean physics; degrades on noisy biology.
“AI can operate in 10,000-dimensional spaces without visualising them”	AlphaFold2 (Jumper et al., <i>Nature</i> , 2021). Also Boltzmann generators (Noé et al., <i>Science</i> , 2019) — learning to sample equilibrium ensembles directly, bypassing long molecular dynamics; and cryoDRGN (Zhong et al., <i>Nature Methods</i> , 2021) for continuous conformational heterogeneity in cryo-EM.
“Formal systems then determine whether the claim is true”	Lean and mathlib ; the Liquid Tensor Experiment (posed Dec 2020, completed Jul 2022).

Table 4. The pasted text’s AI section is the part that most needed citations and had none. With them, it becomes a research programme rather than an aspiration.

Two cautions worth passing to your students

- 1. Prediction is not understanding.** AlphaFold2 predicts structures superbly and explains folding not at all. It is an instrument, in exactly the sense of Move 4 — and like every instrument it generates phenomena that then need a theory. Treating a trained model as if it were a theory is the modern form of Benjamin Peirce’s error.
- 2. The “why?” step has not been automated.** In the *Nature* 2021 work, the machine supplied a *signal*; humans supplied the concept, the conjecture and the proof. The loop currently has a human-shaped hole in the middle of it, and Part VI is a book about what fills that hole.

8 “The next Euler-like synthesis” — assessed

The pasted text closes on an attractive idea: that just as Euler united arithmetic, geometry, trigonometry, calculus and complex numbers, a future synthesis might unite *matter, information, probability, geometry and life*. It is worth taking seriously enough to test — and the test comes from Part VI itself.

What made Euler’s synthesis real

It was **not** a list of fields joined by “+”.

It was a specific claim: two conceptually different *rules* determine the same set of ordered pairs, because both

- map sums to products
- change in proportion to size
- are periodic
- are self-regulating

Four checkable properties. A synthesis is a **shared entailment**, not a shared topic.

What the proposed synthesis currently is

Matter + Information + Probability + Geometry + Life

A list of *subject areas*, with no shared entailment named, no rules compared, and no property that both sides of an equation would have to possess.

This is a research **aspiration**, and a good one. It is not yet a synthesis, and calling it one by drawing a box around it is precisely the move Part VI teaches you to distrust.

Figure 4. The test any claimed synthesis must pass. Euler’s passes it; the proposed one does not yet, and saying so is not a criticism of the ambition — it is a specification of what would have to be produced.

8.1 What a real candidate would look like

The shape of a genuine synthesis in this domain

To count, a proposed unification would have to exhibit **one structure with two conceptually different descriptions that provably entail the same properties**. Candidates that already have this shape, at small scale:

Description A	Description B	Shared entailment
reaction network as a <i>graph</i>	reaction network as a <i>dynamical system</i>	deficiency theorems: graph topology $\Rightarrow$ existence, uniqueness, stability of equilibria
irreversible pulling work	equilibrium free-energy difference	Jarzynski: $\langle e^{-\beta W} \rangle = e^{-\beta \Delta F}$
accuracy of a molecular copier	energy dissipated per step	kinetic proofreading: fidelity is bought with dissipation
precision of a biochemical clock	entropy production rate	thermodynamic uncertainty relation

Each of these *is* a small Euler: a quantity computed two incompatible ways, provably equal. **The realistic path to a large synthesis is to accumulate more of these, not to name the fields you hope will merge.**

9 What to actually do

9.1 Where the original worry is legitimate

I do not want to talk you entirely out of it. A residue is real, and it is worth naming precisely so that it can be worked around rather than worried about.

Residual limit	What it actually costs
A speed and reliability tax	Unintuitive mathematics takes longer to invent, is harder to check, and is taught badly. The parallel postulate cost roughly two millennia, and 39 more years between the symbolic escape and the drawable model. This is an embodiment effect and it is expensive.
Ceilings unrelated to embodiment	Gödel-type incompleteness, and computational intractability, bind any mind. Many-body quantum simulation is exponentially hard for reasons that have nothing to do with how brains work.
A selection effect	We may be systematically failing to notice mathematics we never invented. As stated this is unfalsifiable; hold it lightly, but it is not nothing.

9.2 Four things one person can do

1. Do not try to invent “the mathematics of the cell”

Historically, mathematics that works for a new domain is invented by people forced to compute *one specific blocked thing*.

Fourier
had a heat equation

Heaviside
had a telegraph cable

Feynman
had a QED calculation

Feinberg
had a reaction network

you
need one blocked
calculation

Pick one calculation you cannot currently do, and invent only what it needs. Generality is what you discover afterwards, by stripping the biology out (the pasted text’s step 7 — which is correct, but comes seventh for a reason).

2. Run the metaphor audit as a daily habit

For any formalism you meet, run Part VI’s four questions from p. 436 — what blend makes this true, what relations does it presuppose, what do the symbols mean, where is it grounded — and then add a fifth that belongs to this dossier:

Which grounding in this formalism is the one that will break at $10^{-9} m$?

3. Learn the three languages built for post-intuitive domains

Language	What it replaces	Why it matters here
Category theory	composition, not collection	cells are relational and compositional, not set-like
Sheaves	local data, global obstruction	context-dependent identity — §2.4
A diagrammatic calculus string diagrams, ZX	re-grounds computation in hand and eye	Move 3, the highest-yield move in the history of this problem

4. Treat instruments as mathematics-generators, not just data sources

Every new sensorimotor loop into a new scale has produced new mathematics within roughly a decade. If you are choosing between building a theory for data that exists and building a way to see something nobody has seen, the historical record favours the second more often than mathematicians like to admit.

9.3 The one-sentence answer

You cannot escape embodiment, and you do not need to.

Extend the body with instruments; **re-ground the abstraction** with diagrammatic calculi; **externalise the bookkeeping** to machines.

That three-part move has worked every time it has been tried.

9.4 The question worth giving your students

The pasted text ends by proposing a question for two of your mentees. Their names have been removed here, but the question itself is the best line in the piece and deserves sharpening:

Not

“How much mathematics can you memorise?”

Tests retrieval. Predicts nothing about research capability.

But

“When the mathematics you know fails to explain nature, can you find the assumption that is doing the damage?”

Tests the metaphor audit — Move 1, and the one skill the whole method rests on.

The pasted version asks whether a student can “*invent a better representation.*” That is Move 3, and it is the right ambition — but it is the second step. **You cannot invent a better representation until you have located which embodied assumption the current one is smuggling in.** Newton needed calculus because he had first noticed that the available mathematics could not express instantaneous change; Connes needed non-commutative geometry because he had first noticed that “space is a set of points” was the offending premise.

Teach the diagnosis before the cure.

10 Appendix: dated source ledger

Every substantive claim in this dossier, with its origin. Contested items are marked.

Year	Result	Who / where, and status
Escapes past intuition		
1829–54	Non-Euclidean geometry	Lobachevsky 1829; Bolyai 1832; Riemann 1854
1868	Pseudosphere model (the re-grounding)	Beltrami; later Klein and Poincaré disk models
1932	Hilbert-space formulation of QM	von Neumann, <i>Math. Grundlagen der Quantenmechanik</i>
1936	Non-distributive quantum logic	Birkhoff & von Neumann, <i>Annals of Mathematics</i>
1980s–94	Non-commutative geometry	Connes; <i>Noncommutative Geometry</i> , 1994
2011	Contextuality as a sheaf-theoretic obstruction	Abramsky & Brandenburger, <i>New J. Phys.</i>
Diagrammatic re-grounding (Move 3)		
1948	Feynman diagrams	Feynman
1971	Graphical tensor notation	Penrose
2008–11	ZX-calculus	Coecke & Duncan; complete for qubit QM
Instruments that generated mathematics (Move 4)		
1981	Scanning tunnelling microscope	Binnig & Rohrer; Nobel Prize 1986
1986	Atomic force microscope	Binnig, Quate & Gerber
1978	Forced-unbinding model	Bell, <i>Science</i>
1997	Dynamic force spectroscopy theory	Evans & Ritchie, <i>Biophys. J.</i>
Mathematics that already works for molecular systems		
1972–	Deficiency-zero / deficiency-one theorems	Horn & Jackson 1972; Feinberg onwards
1974–75	Kinetic proofreading	Hopfield (PNAS) 1974; Ninio 1975
1976–77	Stochastic simulation algorithm	Gillespie
1976	Lindblad master equation	Lindblad; Gorini–Kossakowski–Sudarshan same year
1989	Hierarchical equations of motion (HEOM)	Tanimura & Kubo
1997	Non-equilibrium work relation	Jarzynski, <i>Phys. Rev. Lett.</i>
1999	Fluctuation theorem	Crooks, <i>Phys. Rev. E</i>
2002–09	Persistent homology / TDA	Edelsbrunner, Letscher & Zomorodian 2002; Carlsson 2009
2015	Thermodynamic uncertainty relation	Barato & Seifert, <i>Phys. Rev. Lett.</i>
2017	Compositional framework for reaction networks	Baez & Pollard
Machine-extended mathematics (Move 5)		
2019	Boltzmann generators	Noé et al., <i>Science</i>
2020	Symbolic regression for physical law	Udrescu & Tegmark (AI Feynman)
2020–22	Liquid Tensor Experiment	Scholze posed Dec 2020; completed in Lean Jul 2022
2021	AI-guided mathematical intuition	Davies et al., <i>Nature</i> — knot theory and Kazhdan–Lusztig conjectures, both subsequently proved
2021	AlphaFold2	Jumper et al., <i>Nature</i>
2021	cryoDRGN, continuous conformational heterogeneity	Zhong et al., <i>Nature Methods</i>
Quantum biology — graded		
1978	Radical-pair magnetoreception proposed	Schulten et al. STRONG, NOT CLOSED
2000	Decoherence argument against Orch-OR	Tegmark, <i>Phys. Rev. E</i> OBJECTION STANDS
2007	“Long-lived coherence” in light harvesting	Engel et al., <i>Nature</i> SUBSTANTIALLY REVISED
2016	Magnetoreception review	Hore & Mouritsen, <i>Annu. Rev. Biophys.</i>
2021	Cryptochrome-4 magnetic sensitivity <i>in vitro</i>	Xu et al., <i>Nature</i>

10.1 Where to start tomorrow

If you want one entry point into each of the three levers, these are the ones I would pick for a reader with your background:

Lever	Entry point
1 · Body	Single-molecule force spectroscopy: read Bell 1978, then Evans & Ritchie 1997, then Jarzynski 1997. Three short papers, and you will have watched an instrument create a mathematics.
2 · Re-ground	Coecke & Kissinger's diagrammatic treatment of quantum theory. Start with the ZX rewrite rules and prove something small by pushing pictures.
3 · Externalise	Lean and mathlib. Formalise one theorem you already understand — the point is to feel the difference between verified and understood.
The audit	Feinberg's deficiency-zero theorem. It is the cleanest existing example of mathematics built to fit molecular systems rather than borrowed from physics.

Extend the body · Re-ground the abstraction · Externalise the bookkeeping

The embodiment of mathematics is not a ceiling. It is
a specification of where the next lever has to be placed.